Lecture Notes: Shift (Caesar) Ciphers

Math F113X

Terminology

encryption a process of transforming information so that only authorized persons can decode

decryption undoing a transformed message using a set of instructions

plaintext unencrypted message

ciphertext output of decryption

key variable value (word or number) that is needed for encryption and decryption

Shift Cipher (Caesar Cipher)

Each letter in the message is shifted a fixed number of steps over.

Key: what is the shift?

Example: use a shift of 3: $A \to D$, $B \to E$, ...

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
in	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
out	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

1. Encrypt the plaintext E N T R Y

   H Q W U B ← ciphertext

2. Decrypt T X R W H if it was encrypted using a shift of 3.

   Q U O T E ← plaintext